Who Benefits from Cognitive Behavioral Therapy?

Generic Machine Learning Inference on Heterogeneous Treatment Effects

in a Crime-Reduction Experiment in Liberia

Jovitah Namutebi and Pedro Ferreira

ECON 2148: Causal Inference in Economics

May 2026

1 Introduction

In our empirical work, we have personally struggled with the question of studying heterogeneous treatment effects. Conventional approaches usually prescribe pre-specifying a small number of subgroups based on theory or prior evidence, and then testing for differences in treatment effects across these subgroups. Though pre-specification can alleviate concerns about data mining and multiple hypothesis testing, it is still difficult to know in advance which characteristics will be the most relevant moderators of treatment response, especially when rich baseline data is available. Knowing *who* benefits most from an intervention is crucial for policy design and should also inform our understanding of economics.

In this essay, we try to maneuver this challenge by applying the Generic Machine Learning (Generic ML) framework of Chernozhukov et al. (2025), which is relevant for the work of the both of us. Chernozhukov et al. (2025) propose an agnostic, machine-learning-based approach that sidesteps the problems highlighted above by targeting coarser but tractable objects: the Best Linear Predictor of the Conditional Average Treatment Effect (CATE) on an ML proxy, the Sorted Group Average Treatment Effects (GATES), and the Classification Analysis (CLAN). By using repeated sample-splitting and quantile aggregation, their method achieves valid inference under weak assumptions, without requiring the ML estimator to be consistent.

We apply this framework to Blattman et al. (2017)'s randomized experiment, which assigned 999 high-risk men in Monrovia, Liberia, to a 2×2 factorial design crossing an eight-week cognitive behavioral therapy (CBT) program with a $200 cash grant. The experiment is particularly well-suited for Generic ML for three reasons. First, the replication package includes 57 baseline covariates covering demographics, economic status, mental health, personality, time preferences, and criminal history, which can be considered as potential moderators. Second, the randomization was conducted at the individual level, avoiding the small-cluster problem that hampers Generic

ML applications to many field experiments. Third, the sample of $N = 999$ – while modest by ML standards – is among the largest in the literature among trials with individual-level randomization, as far as we know.[1] On top of the methodological convenience, there are two other reasons that make this exercise intellectually interesting. First, a relevant policy question is whether expensive behavioral interventions such as CBT should be universally provided or targeted toward individuals predicted to benefit most. Generic ML provides a natural framework for evaluating whether baseline covariates contain sufficient information to support such targeting. Second, the question of heterogeneity in CBT effects is still open. Blattman et al. (2017) did not pre-specify any heterogeneity analysis, and explicitly did not explore heterogeneity in CBT effects because of that.

We find new results. Using the full factorial sample to maximize power, we find evidence of statistically significant heterogeneity in long-run treatment effects ($\hat{\beta}_2 = 0.557$, $p = 0.060$). The CLAN then characterizes what distinguishes the men at the top of that ranking. A natural prior would be that CBT works best for "marginal" criminals – those with stronger family ties, more social support, lower criminal embeddedness (who face fewer barriers to adopting a prosocial identity). The data suggest the opposite. The men who sustain the largest long-run reductions in antisocial behavior were, at baseline, the most impulsive, the most criminally active, the most psychologically distressed, and the most economically precarious.

The essay proceeds as follows. Section 2 reviews the experimental context. Section 3 describes the Generic ML methodology. Section 4 describes the empirical setup. Section 5 presents and interprets the results. Section 6 concludes.

2 Experimental Context

Blattman et al. (2017) recruited 999 criminally active men in Monrovia, Liberia, aged 18–35, through a targeted street-recruitment process in five high-crime neighborhoods. The sample represents the highest-risk segment of the population: men engaged in drug-selling, theft, and interpersonal violence who would ordinarily be excluded from development programs.

The study used a 2×2 factorial design. In a first lottery, men were assigned to receive or not receive an offer of a CBT program (the Sustainable Transformation of Youth in Liberia, STYL). In a second, independent lottery, men were assigned to receive or not receive a unconditional $200 cash grant. This produced four experimental arms: therapy-only (28%), cash-only (25%), therapy-plus-cash (25%), and pure control (22%).

The CBT intervention. STYL was an eight-week group-based CBT program run by the Network for Empowerment and Progressive Initiatives (NEPI). The curriculum combined psychotherapeutic techniques from two

[1] We considered applying the method to other much larger-scale RCTs, like Banerjee et al. (2023), which comprise virtually millions of individuals. But these are often clustered randomized trials, with no more than hundreds of clusters.

complementary mechanisms. First, it aimed to build self-regulation and patience: facilitators taught men to manage anger, resist impulse, and develop a future-oriented mindset. Second, it promoted a noncriminal social identity: facilitators encouraged men to re-identify with mainstream society, visit banks and markets, and adopt prosocial behaviors. Nearly two-thirds of enrolled men completed the eight-week program.

The cash grant. A separate organization distributed $200 (roughly three months' wages) unconditionally, about three months after therapy ended. The grant was designed as a complementary test of whether financial resources could consolidate the behavioral changes initiated by therapy.

Main findings. Blattman et al. (2017) report large short-term effects of both therapy and cash on antisocial behavior, with therapy alone reducing a standardized antisocial behavior index by roughly 0.20 standard deviations five weeks after treatment. At the twelve-month follow-up, however, only the therapy-plus-cash arm maintains significant reductions in antisocial behavior. The authors hypothesize that cash allowed men to practice the self-control skills acquired in therapy by providing capital for legal self-employment, thereby reinforcing behavior change through learning-by-doing.

3 Generic ML: Quick Overview

We now quickly describe the Generic ML framework following Chernozhukov et al. (2025). The setting is a randomized experiment with N i.i.d. observations (Y_i, D_i, Z_i), where Y_i is the outcome, $D_i \in \{0, 1\}$ is the binary treatment indicator, and Z_i is a (possibly high-dimensional) vector of baseline covariates. The propensity score is known: $p(Z_i) = \Pr(D_i = 1 \mid Z_i)$.

The CATE is the central object of interest:

$$s_0(z) \;:=\; \mathbb{E}[Y(1) - Y(0) \mid Z = z],$$

where $Y(1)$ and $Y(0)$ are potential outcomes. In a randomized experiment, $D \perp (Y(1), Y(0)) \mid Z$, so $s_0(z)$ is identified by

$$s_0(z) = \mathbb{E}[Y \mid D = 1, Z = z] - \mathbb{E}[Y \mid D = 0, Z = z].$$

Because Z may be high-dimensional and s_0 can be a complex nonlinear function, direct nonparametric estimation of s_0 is subject to the curse of dimensionality and, for generic ML estimators, may not converge consistently. Chernozhukov et al. (2025) take an agnostic view: they do not require the ML estimator $S(Z)$ to converge to $s_0(Z)$, and instead build valid inference on coarser features of the CATE.

The approach relies on random sample splitting. The data are partitioned into a main sample M (used for

inference) and an auxiliary sample A (used for ML training). In Stage 1, an ML method is applied to A to produce a proxy predictor $S(Z)$ for the CATE. In Stage 2, $S(Z)$ is used in the main sample to estimate three targets.

To address the randomness of the split, the procedure is repeated over K random splits (here $K = 100$) and results are aggregated via quantile aggregation. Confidence intervals are the median of split-specific confidence bands. Chernozhukov et al. (2025) show that quantile aggregation yields lower estimation risk than a single split.

3.1 Best Linear Predictor (BLP)

The BLP regresses the outcome Y on the proxy $S(Z_i)$ in the main sample, using a modified regression that leverages the Riesz representation:

$$Y_i \;=\; \alpha_0 + \alpha_1 D_i + \underbrace{\beta_1}_{=\text{ATE}} (D_i - p_i) + \underbrace{\beta_2}_{=\text{HET}} (D_i - p_i)[S(Z_i) - \bar{S}] + \varepsilon_i,$$

where $p_i = p(Z_i)$ is the propensity score and $\bar{S}$ is the mean of $S(Z_i)$ in the sample. The coefficient β_1 is a valid estimator of the ATE under weak conditions. Crucially, $\beta_2 > 0$ is the coefficient on the interaction $(D_i - p_i)[S(Z_i) - \bar{S}]$: if the ML proxy contains genuine information about the CATE, then $\beta_2 > 0$. A test of $H_0 : \beta_2 = 0$ is thus a test for whether there is any detectable heterogeneity in treatment effects on the covariates that can be predicted by the ML proxy.

The key theoretical object in the empirical analysis is the BLP coefficient on the interaction between treatment assignment and the ML proxy. Because of this, we include a more formal statement of the BLP result here.

Proposition 1. *Suppose treatment is randomly assigned with known propensity score $p(Z) = P(D = 1 \mid Z)$, where $0 < p(Z) < 1$. Let*

$$s_0(Z) = E[Y(1) - Y(0) \mid Z]$$

denote the conditional average treatment effect, and let $S(Z)$ be a fixed proxy for $s_0(Z)$ obtained from the auxiliary sample. If $\mathrm{Var}(S(Z)) > 0$, then the coefficient β_2 in the population best linear predictor

$$BLP[s_0(Z) \mid S(Z)] = \beta_1 + \beta_2(S(Z) - E[S(Z)])$$

is given by

$$\beta_2 = \frac{\mathrm{Cov}(s_0(Z), S(Z))}{\mathrm{Var}(S(Z))}.$$

Thus, β_2 measures whether the ML proxy ranks individuals according to true treatment-effect heterogeneity. If $\beta_2 = 0$, then the proxy contains no linear predictive information about the CATE.

Proof. See Appendix A.

3.2 Sorted Group Average Treatment Effects (GATES)

GATES sorts individuals into K groups based on their predicted CATE $S(Z_i)$, from the lowest ($k = 1$, least affected) to the highest ($k = K$, most affected). The group average treatment effects $\gamma_k = \mathbb{E}[s_0(Z) \mid G_k]$ are estimated via an auxiliary regression in the main sample. The GATES answer the question:Is the treatment heterogeneous, and in what direction? The contrast $\gamma_4 - \gamma_1$ tests whether the most and least affected groups have different average treatment effects.

3.3 Classification Analysis (CLAN)

CLAN characterizes the most and least affected groups using observable characteristics. For each baseline covariate W_j, CLAN estimates the average of W_j in each GATES group G_k. The contrast $\delta_4^j - \delta_1^j = \mathbb{E}[W_j \mid G_4] - \mathbb{E}[W_j \mid G_1]$ identifies which observable characteristics are associated with being most (vs. least) responsive to treatment. Inference on $\delta_4^j - \delta_1^j$ is valid without multiple-testing correction within each variable, though interpretation across many variables should be cautious.

4 Empirical Setup

We now implement the BLP, GATES, and CLAN estimators described above in the context of Blattman et al. (2017), focusing on the antisocial behavior index and the number of thefts as primary outcomes.

4.1 Treatment Indicators and Outcomes

Because Generic ML requires a binary treatment, we organize our six analyses in two blocks. The first block (A1 through A3) compares the pure CBT treatment arm against the pure control group on three outcomes, short-term and long-term antisocial behavior and short-term thefts. The second block (A4 through A6) exploits the full 2×2 factorial design, pooling both CBT arms as treated ($D = 1$) and both non-CBT arms – pure control and cash-only – as control ($D = 0$), focusing on the same three outcomes. This factorial main-effect estimator uses all $N = 999$ observations in round 1 (or the full long-term sample in A5), roughly doubling the effective sample size relative to A1 through A3 and substantially increasing power for the heterogeneity test. Table 1 summarizes the six analyses.

Table 1: Summary of analyses

Label	Treatment ($D = 1$)	Outcome (Y)	N
A1	CBT only vs. control	Antisocial behavior index (short-term)	463
A2	CBT only vs. control	Antisocial behavior index (long-term)	468
A3	CBT only vs. control	Number of thefts (short-term)	463
A4	CBT (any) vs. cash+control	Antisocial behavior index (short-term)	922
A5	CBT (any) vs. cash+control	Antisocial behavior index (long-term)	947
A6	CBT (any) vs. cash+control	Number of thefts (short-term)	922

Note. In A4–A6, treatment is assigned to any man offered CBT (either CBT-only or CBT+cash); control comprises cash-only and pure-control men. This is the main-effect contrast in a 2×2 factorial design, averaging over cash assignment, which is orthogonal to CBT assignment by construction.

The primary outcome in analyses A1, A2, A4, and A5 is the standardized antisocial behavior index, a composite of stealing, fighting, carrying weapons, drug selling, and hostility. Higher values indicate more antisocial behavior. The short-term outcome is measured approximately five weeks after treatment (round 1 of the panel) while the long-term outcome is measured twelve to thirteen months after treatment (round 5).

Analyses A3 and A6 examine a secondary outcome: the number of thefts committed in the past two weeks. This captures the behavioral outcome most directly linked to crime reduction. We second the discussion around measurement errors and reporting bias to Blattman et al. (2017).

4.2 Covariates

The matrix Z includes all 57 baseline covariates from Blattman's balance table (Table 1 of their paper), all available in their replication package. These include:

- **Demographics**: age, marital status, number of children, schooling, literacy, health, disability.
- **Mental health**: depression and distress scores.
- **Conflict exposure**: war experience, faction membership, ties to military commanders.
- **Economic situation**: weekly earnings, wealth index, homelessness, savings, debt.
- **Labor market**: hours in illicit, agricultural, wage, and high-skill work.
- **Substance use**: drug selling, alcohol use, marijuana use, hard drug use.
- **Criminal history**: number of thefts (violent and non-violent), disputes, hostility.

- **Noncognitive skills**: conscientiousness, neuroticism, grit, reward responsiveness, locus of control, impulsivity, self-esteem.
- **Preferences**: patience (incentivized game and declared), risk aversion, time inconsistency.
- **Cognitive ability**: cognitive test score, executive function.

Missing values in Z (present for a small share of observations) are imputed using column medians. The CLAN is performed on a subset of 19 covariates selected for interpretability.

4.3 Implementation

We use the GenericML R package (Welz and Alfons, 2022), which implements the Chernozhukov et al. (2025) methodology. We set $K = 100$ random splits, with auxiliary proportion 50%, and use 90% confidence intervals throughout. The propensity score is set to constant equal to the sample mean of D (given it is an RCT). Analyses A1, A3 use short-term data (round 1), restricted to the CBT-only arm versus pure control. Analysis A2 uses long-term data (round 5), CBT-only versus pure control. Analyses A4 and A6 use the full round-1 sample ($N = 999$) with the factorial treatment indicator. Analysis A5 uses the full round-5 sample.

5 Results

5.1 Best Linear Predictor: Average Effects and Heterogeneity

Table 2: Best Linear Predictor results

	$\hat{\beta}_1$ (ATE proxy)			$\hat{\beta}_2$ (Heterogeneity)		
Analysis	Estimate	90% CI	p-value	Estimate	90% CI	p-value
A1: CBT only, ST ASB	−0.225	[−0.473, 0.022]	0.073	0.146	[−0.339, 0.654]	0.556
A2: CBT only, LT ASB	−0.010	[−0.250, 0.227]	0.931	−0.244	[−1.258, 0.763]	0.673
A3: CBT only, ST Theft	−0.934	[−2.078, 0.223]	0.109	0.271	[−0.210, 0.861]	0.198
A4: CBT (any), ST ASB	−0.218	[−0.382, −0.054]	0.009	0.282	[−0.158, 0.735]	0.201
A5: CBT (any), LT ASB	−0.176	[−0.334, −0.013]	0.034	0.557	[−0.028, 1.130]	0.060
A6: CBT (any), ST Theft	−0.649	[−1.338, 0.081]	0.078	−0.065	[−0.654, 0.493]	0.819

Notes: 90% confidence intervals from 100 sample splits. $\hat{\beta}_1$ estimates the ATE; $\hat{\beta}_2$ tests for heterogeneity. Best learner: Lasso. Outcome for A1, A2, A4, A5: standardized antisocial behavior index (lower = more prosocial). A3, A6: number of thefts. A4–A6 use the full factorial design (CBT any vs. cash+control).

Average effects ($\hat{\beta}_1$). The BLP results paint a coherent picture across the two blocks of analyses and are broadly consistent with Blattman's ITT estimates. In the narrow comparisons (A1 through A3), CBT alone reduces the standardized antisocial behavior index in the short run ($\hat{\beta}_1 = -0.225$, $p = 0.073$), but this effect appears to dissipate by the twelve-month follow-up ($\hat{\beta}_1 = -0.010$, $p = 0.931$). The effect on the number of thefts is negative and large in magnitude ($\hat{\beta}_1 = -0.934$) but imprecisely estimated ($p = 0.109$), consistent with high variance in self-reported theft counts.

The factorial main-effect analyses (A4 through A6) confirm and considerably sharpen these findings. Using all $N \approx 999$ observations, the estimated CBT effect on short-term antisocial behavior becomes the most precisely estimated result in the table: $\hat{\beta}_1 = -0.218$ ($p = 0.009$). By recovering the marginal effect of CBT averaging over cash assignment (which is orthogonal to CBT by construction), the design doubles the sample and yields a highly significant average effect. The similarity of the point estimates in A1 (-0.225) and A4 (-0.218) suggests that cash assignment does not substantially confound the CBT main effect, as expected from independent randomization.

The factorial design also rescues the long-term finding. In A2, where only CBT-only participants are compared to control ($N = 463$), the twelve-month effect on antisocial behavior is essentially zero ($\hat{\beta}_1 = -0.010$, $p = 0.931$). In A5, which uses the full factorial sample ($N = 950$), the long-term average effect is $\hat{\beta}_1 = -0.176$ ($p = 0.034$) – negative, meaningful, and statistically significant. This suggests that the apparent fade-out of CBT effects visible in the narrow comparison could be partly an artifact of limited power. The factorial effect on short-term theft (A6) is also negative ($\hat{\beta}_1 = -0.649$, $p = 0.078$), marginally significant and consistent with A3.

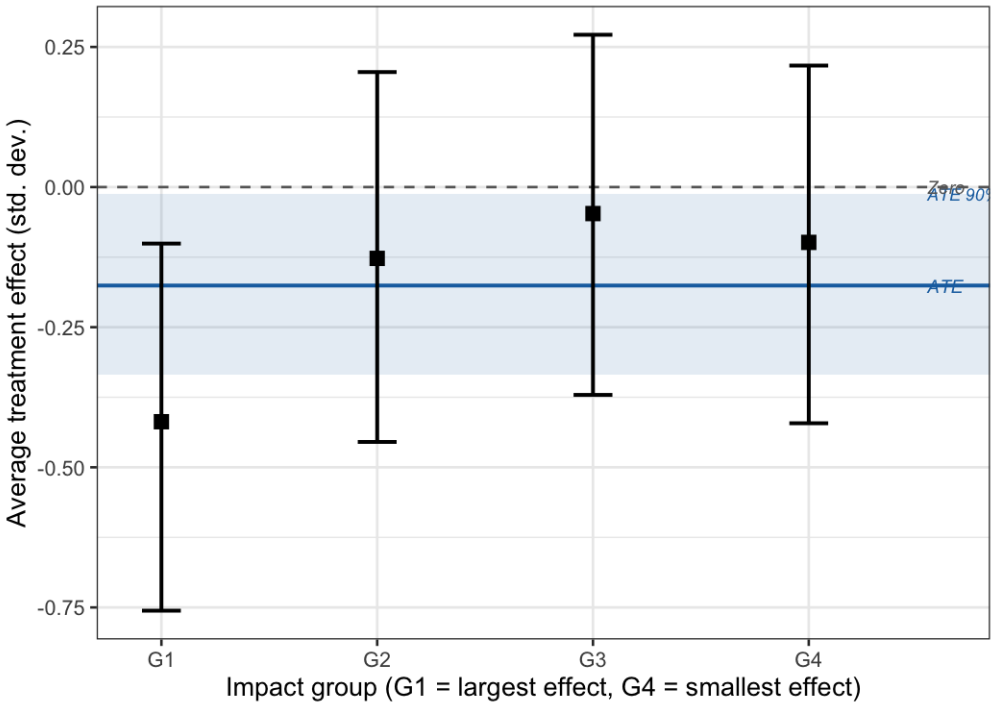
Heterogeneity ($\hat{\beta}_2$). The test cannot reject the null of no heterogeneity in any of the narrow comparisons (A1 through A3). The absence of detectable heterogeneity in A1 through A3 does not imply that treatment effects are uniform. Chernozhukov et al. (2025) note that the power of the BLP test may be limited when true heterogeneity is modest.

However, exploiting the larger sample size of factorial design, the test can reject the null of no heterogeneity in one of the three factorial analyses (A4 through A6). In particular, the most striking result is in A5: the factorial estimate of $\hat{\beta}_2$ for long-term antisocial behavior is 0.557 ($p = 0.060$), statistically significant at the 10% level. This is the only analysis in which Generic ML can reliably predict – from baseline characteristics alone – who benefits most from CBT. The finding has a natural interpretation: while the program shifts the average for most participants in the short run, long-run maintenance of prosocial gains is more selective, concentrated among men whose baseline profiles make them particularly receptive to sustained behavior change. We explore the correlates of this heterogeneity in the GATES and CLAN analyses below.

5.2 Sorted Group Average Treatment Effects (GATES)

We focus the GATES analysis on A5, the analysis that yields significant heterogeneity. Figure 1 plots the four group-average treatment effects for the long-term factorial analysis. Individuals are sorted into quartiles based on the lasso's predicted CATE: G1 has the lowest predicted CATE (and hence the largest actual long-run *reduction* in antisocial behavior) and G4 has the highest predicted CATE (smallest actual reduction). Since treatment reduces antisocial behavior, a more negative γ_k is a larger beneficial effect.

Figure 1: Sorted Group Average Treatment Effects (GATES): CBT (any) vs. cash+control, long-term antisocial behavior (A5, $N = 947$)



Note. Point estimates are medians across 100 splits; bars are 90% confidence intervals. G1 = group with lowest ML-predicted CATE (largest actual effect); G4 = group with highest ML-predicted CATE (smallest actual effect).

G1 shows the largest and most precisely estimated treatment effect: $\gamma_1 = -0.419$ ($p = 0.010$). Groups G2 through G4 are smaller in magnitude and statistically insignificant ($\gamma_2 = -0.127$, $\gamma_3 = -0.047$, $\gamma_4 = -0.099$). The contrast $\gamma_4 - \gamma_1 = 0.350$ ($p = 0.135$) is positive and relatively large, consistent with the significant $\hat{\beta}_2 = 0.557$ ($p = 0.060$) from the BLP.

The pattern is coherent with $\hat{\beta}_2 > 0$ for a negative-direction outcome: a positive $\hat{\beta}_2$ means higher predicted CATE corresponds to a *less negative* actual effect. The men that experience smaller long-run reductions are placed in G4. G1 – with the lowest predicted CATE – concentrates the men with the deepest actual reductions. The spread of effects across the four quartiles ($\gamma_4 - \gamma_1 = 0.350$) quantifies how much the treatment effect varies across the CATE distribution.

5.3 Classification Analysis (CLAN)

Because $\hat{\beta}_2$ is statistically significant in A5, the CLAN for the long-term factorial analysis provides a genuine characterization of who benefits most from CBT in the long run. We focus on A5 as the primary CLAN analysis.

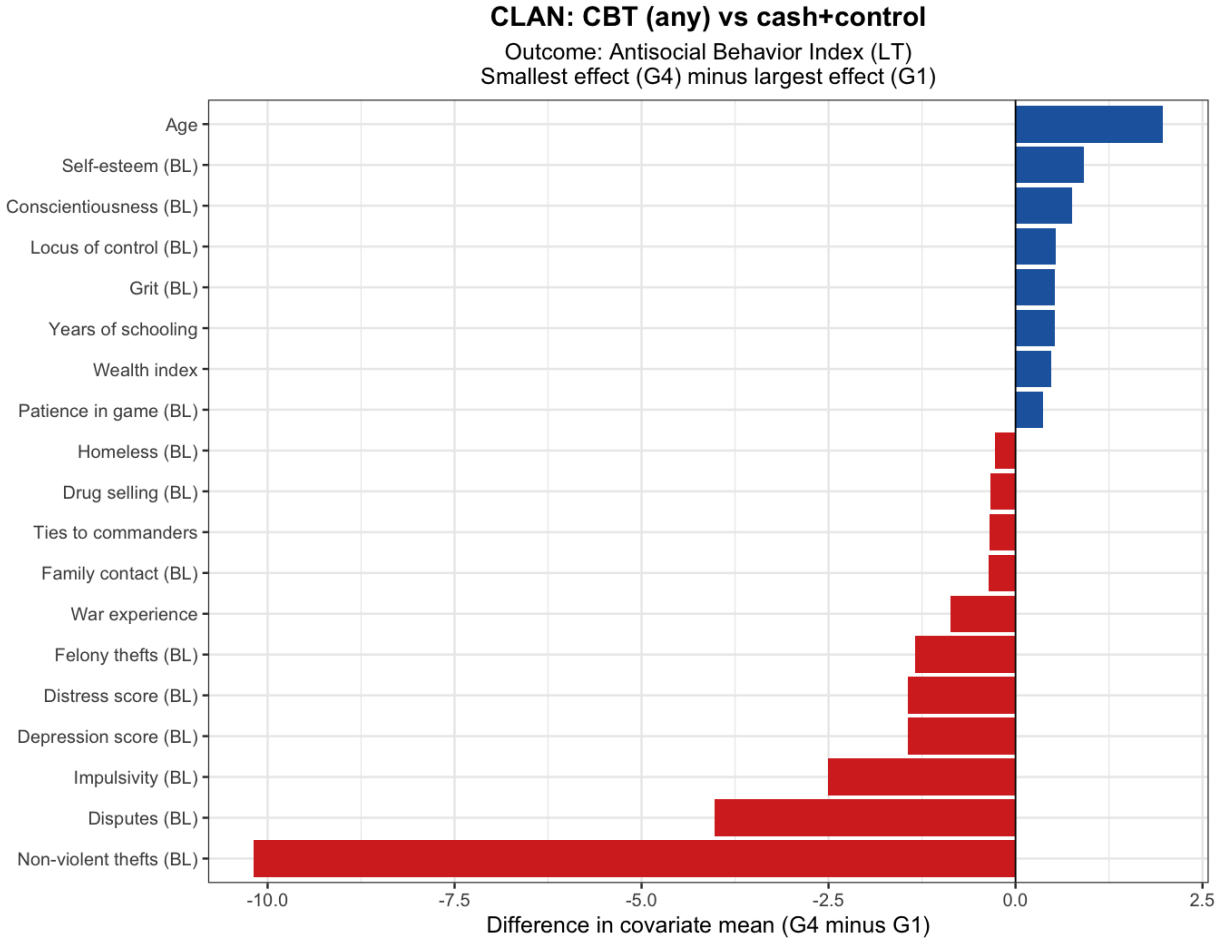


Figure 2: CLAN: CBT (any) vs. cash+control, long-term antisocial behavior (A5). Each bar shows the difference in covariate means between the smallest-effect group (G4, highest predicted CATE) and the largest-effect group (G1, lowest predicted CATE), aggregated as medians over 100 splits (lasso best learner). Positive bars indicate G4 scores higher on that variable; negative bars indicate G1 scores higher.

The contrasts in Figure 2 reveal a sharp and internally consistent profile. The bars show G4 minus G1: positive values mean the smallest-effect group (G4) scores higher; negative values mean the largest-effect group (G1) scores higher. Several contrasts are large and highly significant:

- **Impulsivity**: The smallest-effect group (G4) is 2.59 points *less* impulsive than the largest-effect group (G1) ($\delta_4 - \delta_1 = -2.59$, $p < 0.001$) – the largest contrast in absolute terms. Men with high baseline impulsivity, precisely the psychological profile CBT targets, sustain the largest long-run reductions.

- **Criminal embeddedness**: G4 reports 10.3 fewer non-violent thefts ($p < 0.001$), 1.3 fewer felony thefts ($p < 0.001$), 4.1 fewer disputes ($p < 0.001$), and is 0.33 less likely to have ever sold drugs ($p < 0.001$). G4

also had less war experience (-0.86, $p = 0.017$) and fewer ties to military commanders (-0.36, $p = 0.001$). The largest-effect group (G1) was the most criminally active at baseline.

- **Psychological distress**: G4 scores 1.5 points lower on depression ($p < 0.001$) and 1.45 points lower on psychological distress ($p = 0.006$). The largest-effect group (G1) entered the program more psychologically distressed.

- **Economic precarity**: G4 has a higher wealth index ($+0.48$ standard deviations, $p < 0.001$) and is 0.27 less likely to have been homeless at baseline ($p < 0.001$). The largest-effect group (G1) is poorer and more economically marginalized.

- **Self-regulation**: G4 scores 0.79 points higher on conscientiousness ($p = 0.008$), 0.50 points higher on grit ($p = 0.031$), and 0.93 points higher on self-esteem ($p = 0.003$). The largest-effect group (G1) is less self-regulated with lower self-worth at baseline.

- **Age**: G4 is on average 2.0 years older ($p < 0.001$) and sees family 0.35 fewer days per week ($p = 0.005$).

Taken together, the CLAN identifies the men who sustain the largest long-run reductions in antisocial behavior as those who, at baseline, were the most criminally active, the most psychologically distressed, the most impulsive, the least self-regulated, and the most economically precarious. This is the highest-risk stratum of an already high-risk population.

This finding is counterintuitive. A natural prior would predict that men with more internal resources – lower impulsivity, higher conscientiousness, more stable economic lives – would be best positioned to internalize and sustain the self-regulation lessons of CBT. Instead, the data suggest that it is the men with the most to gain and the deepest behavioral deficits who maintain the largest changes over twelve months. A plausible mechanism is that CBT's identity-transformation component (reidentifying with mainstream society, building self-efficacy, restructuring peer relationships) is most consequential for men whose criminal identity is most salient. For men already more self-regulated and less criminally embedded, the marginal contribution of an eight-week program is smaller.

6 Conclusion

We have applied the Generic ML framework of Chernozhukov et al. (2025) to study who benefits from CBT in Blattman et al. (2017)'s Liberia experiment, organizing six analyses in two blocks: a narrow block (A1–A3) comparing CBT-only against pure control, and a factorial block (A4–A6) exploiting the full 2×2 design to maximize power.

After confirming Blattman et al. (2017)'s main findings on average effects, the central finding is that long-run treatment effects are heterogeneous. While short-term effects appear relatively uniform, in the long-run factorial analysis, the BLP coefficient $\hat{\beta}_2$ is positive and large (0.557) and we can reject the null of no heterogeneity at the 10% level ($p = 0.060$). The CLAN further characterizes the men at the top of the long-run CATE distribution. A natural prior would be that CBT reaches "marginal" criminals – men with stronger family ties, more social support, lower criminal embeddedness – who face fewer barriers to adopting a prosocial identity and are therefore more likely to sustain behavior change. The data point in the opposite direction. The men sustaining the largest long-run reductions in antisocial behavior were, at baseline, the most impulsive, the most criminally active, the most psychologically distressed, and the most economically precarious – the highest-risk stratum of an already high-risk population. This suggests that CBT's identity-transformation mechanism is most consequential precisely where criminal identity is most entrenched, and where the program's social reorientation represents the sharpest departure from baseline conditions.

A Proofs

A.1 Proof of Proposition 1

The BLP is defined as the best linear approximation to the conditional average treatment effect using the proxy $S(Z)$. That is, β_1 and β_2 solve

$$\min_{b_1,b_2} E\left[s_0(Z) - b_1 - b_2(S(Z) - E[S(Z)])\right]^2.$$

Let $\tilde{S}(Z) = S(Z) - E[S(Z)]$. The minimization problem becomes

$$\min_{b_1,b_2} E\left[s_0(Z) - b_1 - b_2\tilde{S}(Z)\right]^2.$$

The first-order condition with respect to b_1 is

$$E\left[s_0(Z) - b_1 - b_2\tilde{S}(Z)\right] = 0.$$

Since $E[\tilde{S}(Z)] = 0$, this implies

$$b_1 = E[s_0(Z)].$$

The F.O.C with respect to b_2 is

$$E\left[\tilde{S}(Z)\{s_0(Z) - b_1 - b_2\tilde{S}(Z)\}\right] = 0.$$

Substituting $b_1 = E[s_0(Z)]$ gives

$$E\left[\tilde{S}(Z)\{s_0(Z) - E[s_0(Z)]\}\right] - b_2E[\tilde{S}(Z)^2] = 0.$$

Therefore,

$$b_2 = \frac{E\left[(S(Z) - E[S(Z)])(s_0(Z) - E[s_0(Z)])\right]}{E\left[(S(Z) - E[S(Z)])^2\right]}.$$

Equivalently,

$$b_2 = \frac{\mathrm{Cov}(s_0(Z), S(Z))}{\mathrm{Var}(S(Z))}.$$

This implies that, the BLP heterogeneity coefficient is the linear projection coefficient of the true CATE on the ML proxy. If this coefficient is zero, then the ML proxy is uncorrelated with the CATE and does not contain linear predictive information about treatment-effect heterogeneity.

■

References

Banerjee, Abhijit, Rema Hanna, Benjamin A Olken, Elan Satriawan, and Sudarno Sumarto, "Electronic food vouchers: Evidence from an at-scale experiment in Indonesia," *American Economic Review*, 2023, *113* (2), 514–547.

Blattman, Christopher, Julian C. Jamison, and Margaret Sheridan, "Reducing Crime and Violence: Experimental Evidence from Cognitive Behavioral Therapy in Liberia," *American Economic Review*, 2017, *107* (4), 1165–1206.

Chernozhukov, Victor, Mert Demirer, Esther Duflo, and Iván Fernández-Val, "Fisher–Schultz Lecture: Generic Machine Learning Inference on Heterogeneous Treatment Effects in Randomized Experiments, with an Application to Immunization in India," *Econometrica*, 2025, *93* (4), 1121–1164.

Welz, Max and Andreas Alfons, "GenericML: Generic Machine Learning Inference," *R package version 0.2.2*, 2022. Available at https://github.com/mwelz/GenericML.